(i) Copy and complete the frequency chart.

Hours	1 – 10	11 – 20	21 – 30	31 – 40
Frequency				

(ii) Draw a bar chart to represent this information.
(iii) How many students use the internet more than 30 hours a week?
(iv) What percentage of students use the internet for 25 hours or more per week?
(v) In what group does the median lie?

8. The table shows the weight loss of people in a slimming club over 6 months.

Weight (w kg)	$0 \le w < 6$	$6 \le w < 12$	$12 \le w < 18$	$18 \le w < 24$	$24 \le w < 30$
Frequency	8	14	19	15	10

(i) Draw a histogram to show these data.
(ii) What is the modal class of the data?

9. This table shows the heights of 60 students in 5th year.

Height (h cm)	$168 \le h < 172$	$172 \le h < 176$	$176 \le h < 180$	$180 \le h < 184$	$184 \le h < 188$	$188 \le h < 192$
Frequency	2	6	17	22	10	3

(i) Draw a histogram of the data
(ii) In which group does the median height lie?
(iii) What is the maximum possible number of students with heights over 182 cm?
(iv) Correct to the nearest cm, what is the greatest possible range of this data?

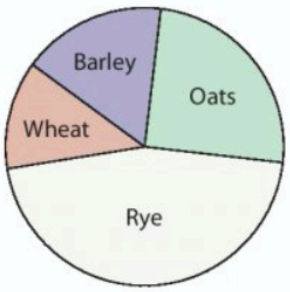
Section 14B Pie Charts

Pie charts are very suitable for categorical data.

All the data is represented by a circle.

Each separate part of the data is a **sector** of the circle.

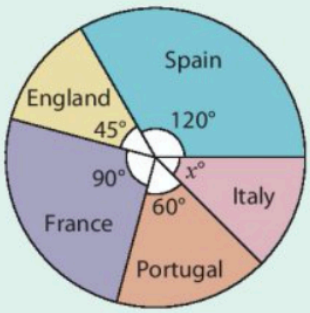
The angle of the sector is proportional to the frequency of that part of the data.



Example 1

On returning to Dublin, 240 holiday makers were asked which countries they had visited. The results are shown in the pie chart on the right.

(i) What is the size of the angle marked $x°$?
(ii) How many people visited Portugal?
(iii) Which country was visited by the most people?
(iv) Which country was visited by $\frac{1}{4}$ of the people questioned?
(v) Which two countries combined accounted for one half of the holiday makers questioned?



(i) The angle at the centre is 360°.

$\therefore 120° + 45° + 90° + 60° + x° = 360°.$

$\Rightarrow \quad 315° + x° = 360°$

$\Rightarrow \quad x = 360° - 315°$

$\Rightarrow \quad x = 45°$

(ii) The angle representing Portugal is 60°.

$\frac{60°}{360°} = \frac{1}{6} \Rightarrow \frac{1}{6}$ of the people visited Portugal

$\frac{1}{6}$ of $240 = \frac{240}{6} = 40$ people

$\therefore$ 40 people visited Portugal.

(iii) 120° is the biggest angle in the pie chart.

Therefore Spain was visited by the most people.

(iv) 90° is $\frac{1}{4}$ of 360° and the angle representing France is 90°.

$\therefore$ France is the country which was visited by $\frac{1}{4}$ of those questioned.

(v) Half of the holiday makers are represented by 180°.

$120° + 60° = 180°$

Therefore Spain and Portugal account for half of the holiday-makers.

Example 2

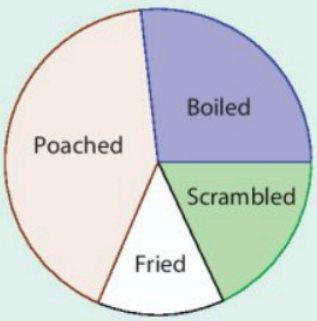
A survey on the different ways students like to eat eggs, produced the following data. Draw a pie chart of this information.

Method of cooking	Fried	Boiled	Scrambled	Poached
Number of students	5	8	6	11

Total number of students $= 5 + 8 + 6 + 11$

$= 30$

Fraction of circle	**Angle**	
$\frac{5}{30} \times 360° =$	60°	Fried
$\frac{8}{30} \times 360° =$	96°	Boiled
$\frac{6}{30} \times 360° =$	72°	Scrambled
$\frac{11}{30} \times 360° =$	132°	Poached



Exercise 14B

1. Simplify the following fractions:

(i) $\frac{12}{36}$ (ii) $\frac{36}{48}$ (iii) $\frac{90}{360}$ (iv) $\frac{120}{360}$ (v) $\frac{9}{18}$

(vi) $\frac{15}{18}$ (vii) $\frac{20}{24}$ (viii) $\frac{45}{360}$ (ix) $\frac{40}{360}$ (x) $\frac{240}{360}$

2. Find the value of each of the following:

(i) $\frac{1}{4}$ of 60 (ii) $\frac{1}{6}$ of 90 (iii) $\frac{1}{3}$ of 84 (iv) $\frac{2}{3}$ of 150 (v) $\frac{1}{8}$ of 176

(vi) $\frac{3}{8}$ of 40 (vii) $\frac{3}{10}$ of 360 (viii) $\frac{1}{12}$ of 360 (ix) $\frac{1}{6}$ of 360 (x) $\frac{2}{3}$ of 360

3. A Biology class produced the following information of the types of trees in a local wood.

Type of tree	Ash	Beech	Oak
Number of trees	20	25	15

Draw a pie chart for this data.

4. A TY class did a colour of eye survey and produced their results in a table format as follows:

Colour of eye	Blue	Brown	Green	Other
Number of students	40	25	15	10

(i) How many students were surveyed?

(ii) Copy and complete the following chart

Colour of eye	Blue	Brown	Green	Other
Number of students	40	25	15	10
Sector angle			$\frac{15}{90} \times 360° = 60°$	

(iii) Draw a pie chart of these results.

5. A Family had a combined income of €60 000. This money was spent according to the pie chart.

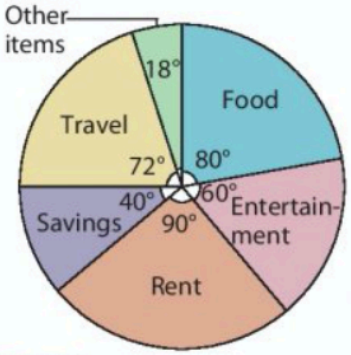
(i) What did they spend most money on?

(ii) What percentage of the income was spent on:

(a) Rent (b) Entertainment?

(iii) How much money was spent on:

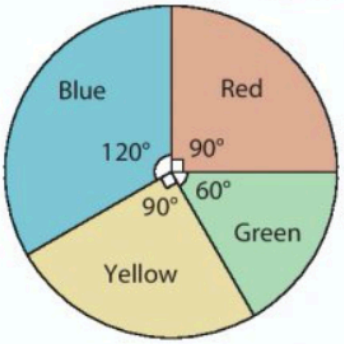
(a) Travel (b) Savings?



6. Sandra drew this pie chart representing the colours of 36 cars in the staff car park.

(i) What fraction of the cars were Green?

(ii) What percentage of the cars were Red?

(iii) How many cars were:

(a) Blue (b) Yellow?